

SD2.0/SDIO2.0/MMC4.1 Host Controller

with AHB Interface



SD Host Spec Ver 2.0 compliant
SDIO Spec Ver 2.0 compliant

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Questions or comments may be directed to:

Arasan Chip Systems, Inc.
1150 North First Street, Suite 210
San Jose, CA 95112
USA

Phone : 408-282-1600
Fax : 408-282-7800
E-mail : sales@arasan.com
Web : <http://www.arasan.com>

Features

- Meets SD Host Controller Standard Specification Draft version 2.0
- Meets SDIO Card Specification version 2.0.
- Meets SD Memory Card Security Specification version 1.01.
- Meets MMC Specification versions 3.31, 4.0, and 4.1
- Supports Test Register to generate events by software
- Supports high capacity Ver2.00 Card
- Supports 64-bit system bus
- Supports MMCPlus™ and MMCMobile™ (versions 3.31, 4.0, and 4.1)
- Supports non-DMA, SDMA, ADMA1, and ADMA2 modes
- Card detection (insertion / removal)
- Password protection of cards
- Host clock rate variable between 0 and 48 MHz
- Supports 1-bit, 4-bit, and 8-bit SD modes
- Supports multi media card interrupt mode.
- Allows card to interrupt host in 1-bit, 4-bit, and 8-bit SD modes
- Up to 10 Mbyte/s read and write rates with 4 parallel data lines (SD 4-bit mode)
- Up to 416 Mbit/s data rate using 8-bit parallel data lines (SD 8-bit mode)
- Cyclic Redundancy Check CRC7 for command and CRC16 for data integrity
- Error Correction Code (ECC) support for MMC4.0 cards
- Designed to work with I/O cards, read-only cards and read/write cards.
- Supports read wait control, suspend/resume operation
- Conforms to AMBA AHB 2.0 specification

Descriptions

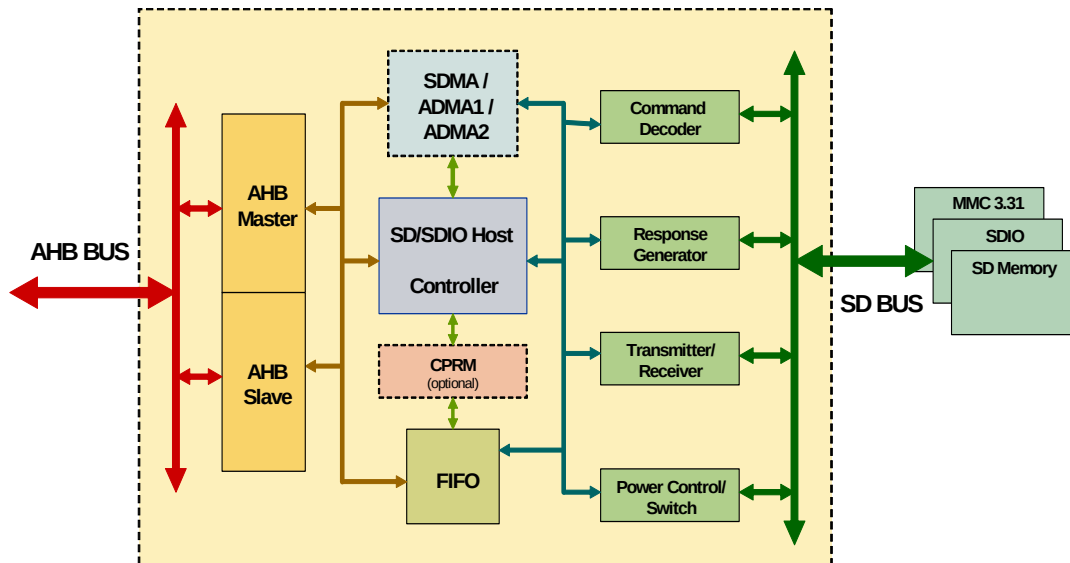
The Arasan SD2.0/SDIO2.0/MMC4.1 Host Controller is a host controller with a ARM processor interface. The product conforms to SD Host Controller Standard Specification Draft Version 2.0.

The SD2.0/SDIO2.0/MMC4.1 Host Controller handles SD2.0/SDIO2.0 protocol at the transmission level including functions such as packing data, adding cyclic redundancy check (CRC), handling start/end bit, and checking transaction format correctness.

The host controller provides programmed IO method and DMA data transfer method. In programmed IO method, the ARM processor transfer data using the Buffer Data Port register. The host controller's support for DMA can be checked by reading the DMA support in the Capabilities register. DMA allows a peripheral to read or write memory without the intervention from the CPU. The system address register points to the first data address, and data is then accessed sequentially from that address.

Functional Block Diagram

SD2.0/SDIO2.0/MMC4.1 - AHB Host Controller Functional Block Diagram



AHB Interface

The Arasan SD2.0/SDIO2.0/MMC4.1 Host Controller provides programmed IO method in which the ARM host driver transfers data using the Buffer Data Port register. The AHB slave consists of the SD2.0/SDIO2.0 control registers and these registers can be programmed by the ARM processor through the AHB slave interface. Data transaction is performed through the AHB slave interface in the case of programmed IO method of data transfer. The AHB Interface initiates a read or write transaction with the memory when data transaction is completed in DMA data transfer mode.

SD2.0/SDIO2.0 Controller

The SD2.0/SDIO2.0/MMC4.1 Host Controller comprises of the AHB host interface, SD2.0/SDIO2.0 controller registers, bus monitor, clock generator, CRC generator and checker (CRC7 and CRC16), and Error Correction Code (ECC) logic.

The AHB host interface acts as the bridge between the AHB host and SD2.0/SDIO2.0/MMC4.1 host controller. The SD2.0/SDIO2.0 controller registers can be programmed by the ARM processor through AHB slave interface. Interrupts are generated to the ARM processor based on values set in the Interrupt status register and interrupt enable register. Bus monitor checks for any violations occurring in the SD bus and time-out conditions. The clock generation block generates SD clock based on the values programmed by the ARM processor in the Clock Control Register. The CRC7 and CRC16 generators calculate the CRC for command and data before sending it to the SD/SDIO card. The CRC7 and CRC16 checker checks for CRC errors of the response and command sent by the SD/SDIO card. To detect data errors on the card, the host may include error correction codes in the payload data. ECC algorithm can be selected to store data on the card. The ECC module can be used by the host or application to decode the user data

SDMA, ADMA1, and ADMA2

The DMA algorithm defined in the SD Host Controller Standard Specification Version 1.00 is called SDMA. Single Operation DMA). This SDMA algorithm forms a performance bottleneck by interruption at every page boundary. ADMA adopts scatter-gather DMA algorithm so that higher data transfer speed is available. ADMA provides data transfer between system memory and SD card without interruption of CPU execution. Support of ADMA can be checked by the Capabilities register.

ADMA can support not only 32-bit system memory addressing but also 64-bit system memory addressing. There are two types of ADMA; ADMA1 and ADMA2. ADMA1 can support data transfer of only 4KB aligned data in system memory. ADMA2 improves the restriction so that data of any location and any size can be transferred in system memory.

DATA FIFO

The SD2.0/SDIO2.0/MMC4.1 Host Controller uses two 1K dual port FIFO for performing both read and write transactions. During a write transaction (data transferred from ARM processor to SD/SDIO card), the data will be filled in to the first and second FIFO alternatively. When data from first FIFO is transferred to the SD card, the second FIFO will be filled and vice versa. The two FIFO's are alternatively used to store data which will give maximum throughput. During a read transaction (data transferred from SD/SDIO card to ARM processor), the data from SD card will be written in to the two FIFO's alternatively. When data from one FIFO is transferred to the ARM processor, the second FIFO will be filled and vice versa and thereby the throughput will be maximized. If the host controller cannot accept data from SD card, it will issue read wait (if card supports read wait mechanism) to stop data from coming in or by stopping the clock.

Note: FIFO depth can be varied using parameter defined in iface_defines.v based on that maximum block size supported will change. If the maximum block size supported is 1K then the FIFO size should be 1K. Two 1K FIFO's are used to support ping-pong mechanism (to increase the throughput).

DAT[0-7] Control Logic

The DAT[0-7] control logic block transmits data through the data line during write transaction and receive data through the data line during read transaction. The interrupt generated from the SD/SDIO card is detected.

Interrupt Controller

The SD2.0/SDIO2.0/MMC4.1 Host Controller generates interrupt to the ARM processor if any of the interrupt bit is set in the interrupt status register.

Command Control Logic

The command control logic block sends commands through the cmd line and receives the response coming from the SD2.0/SDIO2.0/MMC4.1 card.

Power Control

The SD2.0/SDIO2.0/MMC4.1 Host Controller supply SD bus power depending on the value programmed in the Power Control register by the ARM processor. The ARM processor has the responsibility to supply SD bus voltage according to the card OCR and supply voltage capabilities depending on the host controller. If the SD bus power is set to 1 in the Power Control register, the host controller shall supply voltage to the card. If the host driver selects an unsupported voltage in the SD bus voltage select field, the host controller may ignore write to SD bus power and keep its value at zero.

CPRM [Optional]:

CPRM block comprises the Cipher algorithm for encryption and decryption. media unique key, session key, title key are written by the driver to the host controller using cmd60, cmd61, cmd62 respectively. With this session key CCI and title key's are encrypted while performing a secure write transaction and are decrypted while performing a secure read transaction. Command index is kept in track to find out whether authentication process is successful or not. If the command following acmd48 is any one of the command for protected area access, then core assumes that authentication is success and continues with decryption/ encryption for forthcoming security R/W commands. Controller has function calls to en(de)cryption algorithms.

Host Enumeration

The SD2.0/SDIO2.0/MMC4.1 host is enumerated by an external ARM processor. Card insertion or removal in the slot is informed to the processor via interrupt methodology. The cards in the slot are enumerated by the SD host controller as instructed by the processor via slave register sets.

Signal Interfaces

Legend: I = Input only; O = Output only; I/O = Input/Output.

Table 1: AHB Master / Slave interface

Signal Name	I/O	Description
clk_ahb	I	AHB System Clock.
m_hbusreq	O	AHB Bus request
m_hgrant	I	AHB Bus Grant
m_haddr[31:0]	O	DWord Address
m_hwdata[31:0]	O	AHB master write data
m_hrdata[31:0]	I	Read data
m_hwrite	O	Write/Read Direction Indication
m_hsize[1:0]	O	Size (byte, half word or word)
m_hburst[2:0]	O	Burst Size
m_hready	I	Ready signal
m_htrans[1:0]	O	Transfer type
m_hresp[1:0]	I	Transfer response
int_to_arm	O	Interrupt to the ARM
t_hsel	I	Slave Select
t_haddr[5:0]	I	DWord Address (256 bytes)
t_hwdata[31:0]	I	Write Data
t_hrdata[31:0]	O	Read Data
t_hwrite	I	Write / Read Direction Indication
t_hsize[1:0]	I	Size (Byte, Half Word or Word)
t_htrans[1:0]	I	Transfer Type
t_hready	O	Slave Ready
t_hresp[1:0]	O	Transfer Response

Table 2: System Interface

Signal Name	I/O	Description
clk_xin	I	Crystal Oscillator Clock. clk_sdcard is derived from this clock. This clock is also used to derive 32Khz clock which is used to detect Card insertion / removal. For maximum efficiency this should be around 50Mhz.

Table 3: SD/SDIO card Interface

Signal Name	I/O	Description
clk_xin	I	Crystal Oscillator Clock. clk_sdcard is derived from this clock. This clock is also used to derive 32Khz clock which is used to detect Card insertion / removal. For maximum efficiency this should be around 50Mhz.
clk_sdcard	O	SDIO/SD Clock
cmd_di	I/O	SD4 bit mode: Command Line SD1 bit mode: Command Line
data0	I/O	SD4, SD8 bit mode: Data Line 0 SD1 bit mode: Data Line
data1	I/O	SD4, SD8 bit mode: Data Line1 or Interrupt (optional) SD1 bit mode: Interrupt
data2	I/O	SD4, SD8 bit mode: Data Line2 or Read Wait (optional) SD1 bit mode: Read Wait (optional)
data3	I/O	SD4, SD8 bit mode: Data Line 3 SD1 bit mode: Not Used
data4	I/O	SD8 bit mode: Data Line 4 SD1 / SD4 bit mode: Not Used
data5	I/O	SD8 bit mode: Data Line5 SD1 / SD4 bit mode: Not Used
data6	I/O	SD8 bit mode: Data Line6 SD1 / SD4 bit mode: Not Used
Dat6_s1_in	I	
Dat6_s1_out	O	
Dat6_s1_out_en	O	
data7	I/O	SD8 bit mode: Data Line 7 SD1 / SD4 bit mode: Not Used
sdcd_n	I	Card Detection for single Slot
sdwp_n	I	SD Card Write Protect
led_on	O	LED ON: To Caution the user not to remove the card while the SD card is being accessed

Table 4: Power Supply Controller

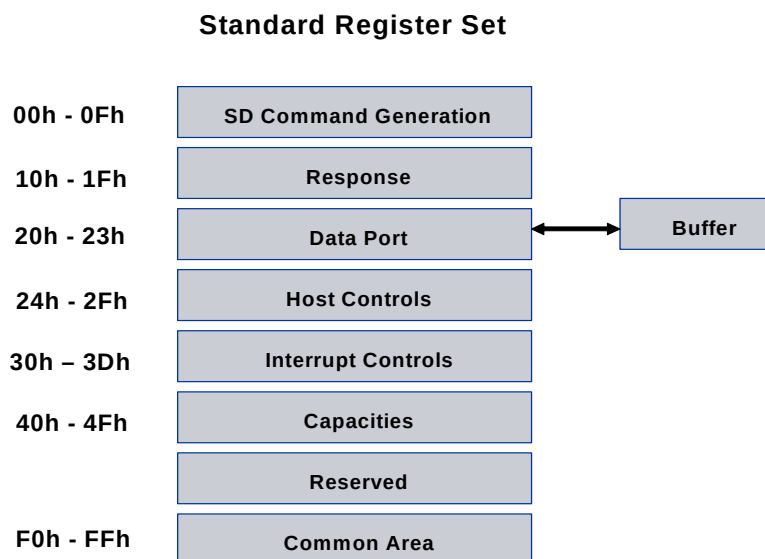
Signal Name	I/O	Description
bus_pow	O	Control SD Card Power Supply
bus_volt[2:0]	O	SD Bus Voltage select.

Table 5: SD2.0 Host Controller Register Map

* Color code for slot 1 and slot 2: - SLOT 1 - SLOT 2

Offset	15-08 bit	07-00 bit	Offset	15-08 bit	07-00 bit
002h	System Address(High)		000h	System Address (Low)	
006h	Block Count		004h	Block Size	
00Ah	Argument1		008h	Argument0	
00Eh	Command		00Ch	Transfer Mode	
012h	Response1		010h	Response0	
016h	Response3		014h	Response2	
01Ah	Response5		018h	Response4	
01Eh	Response7		01Ch	Response6	
022h	Buffer Data Port1		020h	Buffer Data Port 0	
026h	Present State		024h	Present State	
02Ah	Walk-up Control	Block Gap Control	028h	Power control	Host Control
02Eh	Software Reset	Time-out control	02Ch	Clock Control	
032h	Error Interrupt Status		030h	Normal Interrupt Status	
036h	Error Interrupt Status Enable		034h	Normal Interrupt Status Enable	
03Ah	Error Interrupt Signal Enable		038h	Normal Interrupt Signal Enable	
03Eh	---		03Ch	Auto CMD12 Error Status	
042h	Capabilities		040h	Capabilities	
046h	Capabilities (Reserved)		044h	Capabilities (Reserved)	
04Ah	Maximum Current Capabilities		048h	Maximum Current Capabilities	
04Eh	Maximum Current Capabilities (Reserved)		04ch	Maximum Current Capabilities (Reserved)	
---	---		---	---	
0F2h	---		0F0h	---	
---	---		---	---	
0FEH	Host Controller Version		0FCh	Slot Interrupt Status	

Classification of the standard register map



Multi-Slot Support:

One Standard register is defined for each slot. If the Host Controller has two slots, two register sets shall be required. Each slot is controlled independently. This enables support for combinations of full-speed and low-speed cards in regards to SD clock frequencies.

Verification

IP Proven on

- Altera FPGA based platform
- Silicon Proven - on customer silicon's.

Licensees include TOSHIBA, HP, LSI Logic nVidia, ATI and over 20 other customers.

Deliverables

The full IP package complete with:

- Verilog HDL
- Synthesis scripts
- Test Environment and test scripts
- SD2.0/SDIO2.0 Host User guide with full programming interface and parameterization instructions.
- FPGA prototype Board - On request

Arasan Chip Systems Inc. is an Executive Member of the SD Card Association since 2001 and participates in the SD/SDIO Standards and Compliance Workgroups.